

Impact of Image Enhancement and Augmentation on Skin Lesion Classification: An Experimental Comparison of Classical Machine Learning and Deep Learning Models

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Abstract—Automated dermoscopic image classification is challenging because visually similar lesions may belong to different diagnoses and medical-image datasets are often strongly imbalanced. This project compares Logistic Regression, Random Forest, a Long Short-Term Memory network (LSTM), a custom convolutional neural network (CNN), and ImageNet-pretrained EfficientNetB0 on the seven-class HAM10000 dataset. A majority-class classifier is included as a reference baseline. The dataset contains 10,015 images from 7,470 unique lesions. To reduce leakage, train, validation, and test subsets are separated by lesion identifier using stratified grouped folds. Balanced accuracy and macro F1 are emphasized because melanocytic nevi dominate the dataset. A controlled 2×2 ablation is then performed for CNN and EfficientNetB0: (A) baseline, (B) black-hat hair removal plus CLAHE only, (C) online augmentation only, and (D) enhancement plus augmentation. In the controlled baseline, the CNN achieves the highest balanced accuracy, 0.6775, while EfficientNetB0 achieves higher accuracy and macro F1. Augmentation-only EfficientNetB0 gives the strongest overall accuracy, macro F1, and log loss at 0.7345, 0.5848, and 0.7550, respectively. Enhancement does not improve overall deep-model performance, and combining enhancement with augmentation reduces balanced accuracy for both neural models. These results show that preprocessing choices are architecture-dependent and should be validated rather than assumed to help.

Index Terms—machine learning, skin lesion, HAM10000, convolutional neural network, EfficientNet, transfer learning, class imbalance, data augmentation, CLAHE

I. INTRODUCTION

Medical condition classifications are prominently challenging problems, especially skin lesion classification from dermoscopic images. Different skin-lesion diagnoses can share color, texture, border, and shape characteristics, while lesions belonging to the same diagnosis can vary substantially. HAM10000 is additionally imbalanced: the melanocytic-nevus class (*nv*) accounts for roughly two thirds of the data, whereas dermatofibroma (*df*) and vascular lesions (*vasc*) are rare. Under this distribution, ordinary accuracy can give an overly favorable impression of a model that mainly predicts the dominant class.

This project evaluates seven-class classification of actinic keratoses and intraepithelial carcinoma (*akiec*), basal cell

carcinoma (*bcc*), benign keratosis-like lesions (*bkl*), dermatofibroma (*df*), melanoma (*mel*), melanocytic nevi (*nv*), and vascular lesions (*vasc*) [1]. The study compares linear, tree-based, recurrent, scratch-convolutional, and transfer-learning approaches. The final stage is a controlled ablation designed to answer a narrower question: do image enhancement such as black-hat hair removal, CLAHE contrast enhancement, and online augmentation always improve CNN and EfficientNetB0 performance when all other settings are held fixed?

The main contributions are: (1) lesion-aware train/validation/test splitting to reduce leakage; (2) class-weighted training and class-balanced metrics; (3) comparison of classical, recurrent, scratch-CNN, and transfer-learning models; and (4) a 2×2 preprocessing ablation that separates enhancement from augmentation.

II. RELATED WORK AND BACKGROUND

HAM10000 was introduced as a multi-source dermoscopic dataset for research on common pigmented skin lesions [1]. Logistic Regression provides a linear reference over flattened pixels, while Random Forest can model nonlinear interactions through an ensemble of randomized decision trees [2]. LSTM networks use gated recurrent memory [3]; although designed for sequential data, an image can be reshaped into a sequence of rows for comparison. CNNs preserve two-dimensional spatial structure and learn hierarchical local features. Neural models are optimized with Adam [4]. EfficientNetB0 uses compound scaling of depth, width, and input resolution and is used here with ImageNet transfer learning [5].

Image preprocessing is often proposed as a way to improve dermoscopic classification. Hair can obscure lesion structures, while local contrast enhancement may emphasize boundaries and texture. In this project, black-hat morphology followed by inpainting is used for hair suppression, and CLAHE is applied to the luminance channel. Separately, online augmentation introduces random flips, rotations, small translations and scaling, and mild color perturbations during training. Because these transformations may remove, distort, or regularize different visual cues, their benefit is treated as an empirical question.

III. DATASET AND EXPERIMENTAL DESIGN

A. Dataset and Leakage-Safe Split

The dataset contains 10,015 images from 7,470 unique lesions. Labels are encoded as seven diagnosis categories. A ten-fold StratifiedGroupKFold split with random seed 42 is grouped by `lesion_id`; fold 0 is used for testing, fold 1 for validation, and folds 2–9 for training. The resulting split contains 8,011 training images, 1,002 validation images, and 1,002 test images. Assertions verify that no lesion identifier appears in more than one subset.

Brightness and contrast outliers are detected from the training data only using interquartile-range thresholds. Seventy training images are removed, leaving 7,941 training images; validation and test data are unchanged. The test set contains 32 *akiec*, 51 *bcc*, 110 *bkl*, 12 *df*, 112 *mel*, 671 *nv*, and 14 *vasc* images. Thus, *nv* alone accounts for approximately 67% of the test set whereas *df* and *vasc* each contribute roughly 1%.

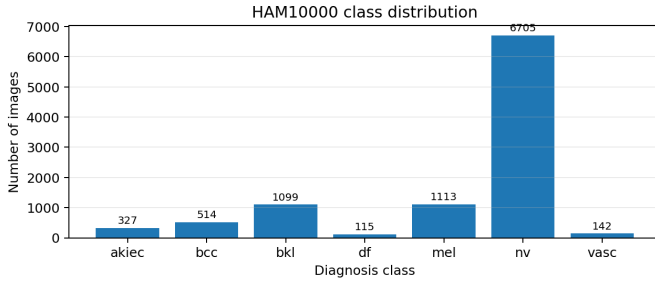


Fig. 1. Class distribution in HAM10000, illustrating the strong dominance of the *nv* class.

B. Input Representations

Logistic Regression, Random Forest, and LSTM use RGB images resized to 32×32 and normalized to $[0, 1]$. Logistic Regression and Random Forest flatten each image to 3,072 pixel features. The LSTM reshapes each image into 32 time steps with 96 RGB values per step. The CNN and EfficientNetB0 use 224×224 RGB inputs.

Class weights are computed from the filtered training labels. Neural models minimize weighted multiclass cross-entropy,

$$\mathcal{L} = -\frac{1}{n} \sum_{i=1}^n w_{y_i} \log p(y_i | x_i). \quad (1)$$

Because the data are imbalanced, balanced accuracy is a primary metric,

$$\text{BA} = \frac{1}{7} \sum_{k=1}^7 \text{Recall}_k. \quad (2)$$

Macro precision, macro recall, macro F1, ordinary accuracy, and log loss are also reported.

C. Four-Experiment Neural Ablation

The final ablation changes only two factors for CNN and EfficientNetB0. Experiment A uses original images and no random augmentation. Experiment B applies black-hat hair removal and CLAHE but no random augmentation. Experiment C uses original images with online augmentation. Experiment D combines enhancement and online augmentation. Validation and test data are never enhanced and randomly augmented.

For CNN, augmentation consists of horizontal and vertical flips, rotations up to $\pm 20^\circ$, small affine translation and scaling, and mild brightness, contrast, saturation, and hue jitter. EfficientNetB0 uses comparable Keras augmentation layers. Enhancement is deterministic and therefore cached; augmentation is generated online during training. All four neural experiments use the same lesion split, retained training rows, image sizes, architecture, class weights, optimizer settings, epoch budget, random seed, and model-selection criteria.

IV. MODELS

A. Classical and Recurrent Models

The majority baseline always predicts the dominant class which is *nv*. Logistic Regression is trained on flattened $32 \times 32 \times 3$ inputs using the SAGA solver, L2 regularization with $C = 1.0$, tolerance 10^{-3} , a maximum of 500 iterations, and balanced class weights. Random Forest uses 100 trees, random seed 42, all available CPU cores, and balanced class weights.

The LSTM processes each $32 \times 32 \times 3$ image as 32 time steps of 96 features. It contains one 64-unit LSTM layer, dropout of 0.3, a 64-unit ReLU dense layer, and a seven-unit softmax output. It is trained with Adam and sparse categorical cross-entropy.

B. Custom CNN

The custom compact VGG-style architecture CNN trained from scratch contains three convolutional blocks with channels 32, 64, and 128. Each block contains two 3×3 convolutions, batch normalization, ReLU, and 2×2 max pooling. A 1×1 adaptive global average pooling layer reduces the final feature maps to 128 values. The classifier contains a 256-unit ReLU layer, dropout of 0.5, and a seven-logit output layer. The model contains 322,727 trainable parameters. It is optimized with Adam at learning rate 0.001, cosine annealing, weighted cross-entropy, and validation balanced-accuracy checkpointing.

C. EfficientNetB0

EfficientNetB0 is initialized with ImageNet weights. In stage 1, the pretrained backbone is frozen while a new global-average-pooling and dropout classification head is trained. In stage 2, a portion of the backbone is unfrozen for fine-tuning at a smaller learning rate. Class weights are used in both stages. The selected checkpoint is determined from validation performance, and the held-out test set is used only for final evaluation.

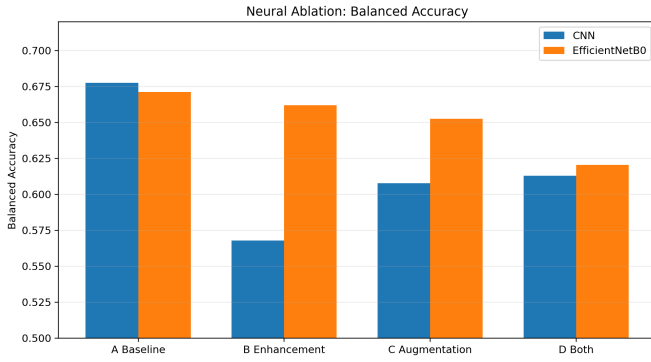


Fig. 2. Balanced accuracy across the four controlled neural experiments.

V. RESULTS

A. Controlled Baseline Comparison

Table I reports Experiment A, which provides the common no-enhancement/no-augmentation reference. The majority baseline demonstrates the weakness of raw accuracy: it achieves 0.6697 accuracy but only 0.1429 balanced accuracy and 0.1146 macro F1 because it predicts only *nv*.

The CNN obtains the highest balanced accuracy in the controlled baseline, 0.6775, indicating strong average recall across all seven diagnoses. EfficientNetB0 obtains substantially higher ordinary accuracy, macro precision, macro F1, and better log loss. The CNN therefore favors class-wide recall more strongly, while EfficientNetB0 provides a better precision–recall balance overall.

The CNN baseline confusion matrix shows recalls of approximately 0.72 for *akiec*, 0.75 for *bcc*, 0.58 for *df*, and 0.93 for *vasc*, but only 0.65 for the dominant *nv* class. This explains how balanced accuracy can exceed ordinary accuracy: the class-weighted CNN sacrifices some majority-class accuracy in exchange for stronger minority-class recall.

B. Four-Experiment Ablation

Table II summarizes the controlled neural ablation. No single configuration dominates every metric.

For EfficientNetB0, augmentation alone (C) is the strongest overall configuration. Relative to A, accuracy increases from 0.7146 to 0.7345 and macro F1 from 0.5536 to 0.5848, while log loss improves from 0.8107 to 0.7550. Balanced accuracy decreases modestly from 0.6712 to 0.6526. Thus, augmentation improves overall precision–recall balance and probability quality but slightly reduces average per-class recall. The confusion matrix indicates that augmentation improves recall for *bkl*, *mel*, and *nv*, while recall falls for *akiec*, *bcc*, and *df*.

Enhancement alone (B) does not provide an overall benefit for EfficientNetB0. Balanced accuracy changes only slightly, from 0.6712 to 0.6620, but accuracy falls to 0.6776, macro F1 to 0.5080, and log loss worsens to 0.8907. Combining

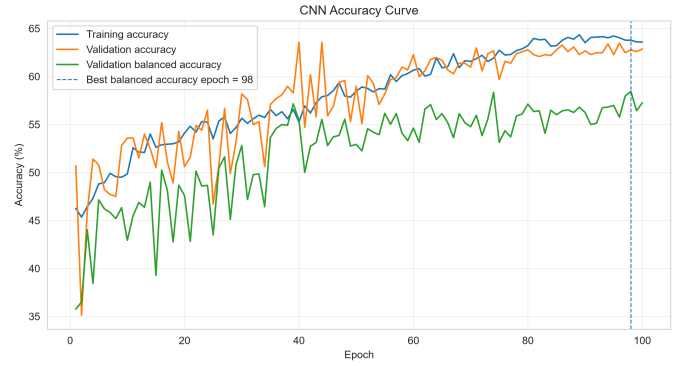


Fig. 3. CNN baseline training accuracy, validation accuracy, and validation balanced accuracy. The selected checkpoint is based on validation balanced accuracy.

enhancement with augmentation (D) is also worse than augmentation alone: balanced accuracy falls to 0.6206 and macro F1 to 0.5176. The evidence therefore suggests that black-hat hair suppression plus CLAHE does not complement the pretrained EfficientNet features in this configuration.

For the scratch CNN, the baseline (A) is the strongest configuration by balanced accuracy and macro F1. Enhancement alone reduces balanced accuracy by about 11 percentage points, from 0.6775 to 0.5678. Augmentation alone also reduces balanced accuracy, to 0.6077, and macro F1, to 0.4657. The combined configuration partially recovers performance, reaching 0.6130 balanced accuracy and 0.4932 macro F1, but still does not surpass the baseline. These results show that regularization and image enhancement are not automatically beneficial for a small scratch CNN.

TABLE I
EXPERIMENT A BASELINE COMPARISON ON THE HELD-OUT TEST SET. HIGHER IS BETTER EXCEPT FOR LOG LOSS.

Model	Accuracy	Bal. Acc.	Macro P	Macro R	Macro F1	Log Loss
EfficientNetB0	0.7146	0.6712	0.4968	0.6712	0.5536	0.8107
CNN	0.6397	0.6775	0.4453	0.6775	0.5088	0.9419
Random Forest	0.6597	0.4051	0.3914	0.4051	0.3871	0.9001
LSTM	0.5150	0.5265	0.3487	0.5265	0.3722	1.1268
Logistic Regression	0.5569	0.4805	0.3285	0.4805	0.3535	1.3619
Majority Baseline	0.6697	0.1429	0.0957	0.1429	0.1146	11.9066

TABLE II
CONTROLLED CNN AND EFFICIENTNETB0 ABLATION. A: BASELINE; B: ENHANCEMENT ONLY; C: AUGMENTATION ONLY; D: ENHANCEMENT + AUGMENTATION.

Exp.	Model	Accuracy	Bal. Acc.	Macro F1	Log Loss
A	EfficientNetB0	0.7146	0.6712	0.5536	0.8107
A	CNN	0.6397	0.6775	0.5088	0.9419
B	EfficientNetB0	0.6776	0.6620	0.5080	0.8907
B	CNN	0.6467	0.5678	0.4592	0.9011
C	EfficientNetB0	0.7345	0.6526	0.5848	0.7550
C	CNN	0.6108	0.6077	0.4657	0.9756
D	EfficientNetB0	0.7136	0.6206	0.5176	0.8089
D	CNN	0.6387	0.6130	0.4932	0.9035

VI. DISCUSSION

The four-experiment design provides stronger evidence than a single enhancement-plus-augmentation run because it separates the two image interventions. For EfficientNetB0, the pattern is clear: augmentation alone improves accuracy, macro F1, and log loss, whereas deterministic enhancement lowers overall performance. For CNN, neither intervention improves the baseline. The interaction is also dependent on architectures: combining enhancement and augmentation is worse than augmentation alone for EfficientNetB0, but it recovers some of the CNN performance lost under either intervention separately.

Metric choice changes which model appears best. If the primary objective is equal average recall across diagnoses, the baseline CNN is strongest with 0.6775 balanced accuracy. If the objective is a stronger overall balance of precision, recall, probability quality, and raw accuracy, augmentation-only EfficientNetB0 is preferable, with 0.7345 accuracy, 0.5848 macro F1, and 0.7550 log loss. This distinction is important in an imbalanced medical-image task: a single metric does not fully describe model behavior.

The experiments also illustrate the limits of preprocessing assumptions. Black-hat filtering may remove hair, but inpainting can modify dark lesion structures. CLAHE increases local contrast but may amplify texture that is not class-discriminative. A pretrained network may already contain robust low-level filters, so aggressive deterministic preprocessing can shift the input distribution away from the natural images represented during pretraining. In contrast, online augmentation does not permanently alter every sample and can expose the model to a broader range of plausible appearances.

VII. LIMITATIONS

Several limitations remain. First, the rarest test classes contain only 12 *df* and 14 *vasc* images, so a few predictions can cause large changes in per-class recall. Second, each deep-learning configuration was trained once; repeated seeds or repeated grouped folds would be required to estimate variance and determine whether small differences are statistically stable. Third, black-hat filtering and CLAHE are grouped together as one enhancement treatment, so this experiment does not evaluate their individual effects. Fourth, the classical models and the LSTM are not included in the four-experiment analysis as they are reported from the common baseline and only serve as reference points for the model-family. Finally, performance on HAM10000 does not establish clinical diagnostic effectiveness and would require independent external validation before any clinical interpretation.

VIII. CONCLUSION AND FUTURE WORK

This project compares classical machine learning, recurrent modeling, a custom CNN, and EfficientNetB0 transfer learning for seven-class HAM10000 skin-lesion classification under a lesion-aware and class-imbalanced evaluation protocol. The controlled baseline shows that the scratch CNN achieves the highest balanced accuracy, 0.6775, whereas EfficientNetB0 achieves higher accuracy, macro F1, and better log loss. The majority baseline confirms that raw accuracy alone is misleading, since it achieves 0.6697 accuracy while recognizing only one class.

The 2×2 ablation gives a more specific conclusion. For EfficientNetB0, online augmentation without image enhancement produces the strongest overall result: 0.7345 accuracy, 0.6526 balanced accuracy, 0.5848 macro F1, and 0.7550 log loss.

Black-hat and CLAHE does not improve the pretrained model and weakens the augmentation-only result when combined with it. For the custom CNN, the unenhanced, unaugmented baseline remains best in balanced accuracy and macro F1. The broader conclusion is that augmentation and enhancement are architecture-dependent design choices that must be evaluated empirically rather than assumed to improve performance.

Future work could be repeating the four configurations across multiple random seeds or grouped folds, separate black-hat filtering from CLAHE in a finer ablation, tune augmentation strength on validation balanced accuracy and macro F1, evaluate focal loss or balanced sampling, and compare additional pretrained backbones. These extensions would clarify whether the observed effects are stable and may improve rare-class performance.

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